

	31	32	
3ai	The gravitational potential at a point is the <u>work done per unit mass</u> in bringing a <u>small test mass</u> from <u>infinity</u> to that point.		B1 B1
3aii	Gravitational potential is always negative because the gravitational potential is taken to be <u>zero at infinity</u>		B1

	and gravitational forces are <u>attractive</u>, <u>work done by the external agent</u> on the point mass moving it from infinity is <u>negative</u>. OR (test) mass <u>getting closer</u> (from infinity) <u>loses potential energy</u>.	B1 B1
3b	 one curve with negative field strength near planet with gradient of decreasing magnitude at $2R$ and finishing with positive field strength near the moon with gradient of increasing magnitude at $D - R$ field strength shown as zero (only) near the point of maximum potential (within the 4th big square from left, acceptable range of 4th to 8th small square).	B1 B1
3ci	At Earth's surface, $g = \frac{GM}{R_E^2}$ $\rightarrow gR_E^2 = GM$ where R_E is radius and M is mass of Earth gravitational force on Moon, $F = \frac{GMm}{r^2} = \frac{gR_E^2 m}{r^2}$ $= \frac{9.81(6.37 \times 10^6)^2 (7.35 \times 10^{22})}{(3.84 \times 10^8)^2}$ $= 1.9841 \times 10^{20}$ $= 1.98 \times 10^{20} \text{ N}$	M1 M1 A0
3cii	Since the increase in orbit is significantly smaller than radius of orbit, the gravitational force remains constant. Hence, change in potential energy = $F \times 0.040$ $= 1.98 \times 10^{20} \times 0.040$ $= 7.9 \times 10^{18} \text{ J}$ do not accept alternative method of finding change in p.e. if not using answer in (c)(i)	B1 C1 A1
4a	thermal energy per unit mass required to produce unit rise of temperature of the	B1